

Lower Your Cloud Bill 101:

An Intro to ~~Rate Optimization~~

~~Lowering Cloud Spend Through Rate Optimization~~

What is Rate

Optimization anyway?

The discipline of lowering the unit cost of a particular cloud service through ~~pricing~~ discounts.

Rate Optimization vs. Usage Optimization

- | | |
|----------------------------------|------------------|
| - Reserved Instances | - Idle Resources |
| - Savings Plans / Flexible UDS | - Right sizing |
| - Spot instances | - Autoscaling |
| - EDC (Enterprise Discount Plan) | - Scheduling |
| - Negotiated SKU discounts | |
| - Partner Discounts | |

* The most powerful tool to lower your cloud bill

Right Away → long-term commitments

Long-term Commitments: An agreement between you and a cloud service provider (CSP) for dedicated usage in exchange for pricing discounts.

Reserved Instances

vs.

Savings Plans

- | | |
|--|-------------------------------|
| - Traditional RIs, Committed-use discounts (UDS) | - Savings Plans, Flexible UDS |
| - Resource, Region, Machine Type | - Dedicated H spend |
| - 1 to 3 years | - 1 to 3 years |
| - up to 72% off list | - up to 66% off list |

Understanding Your Options: Types of Cloud Commitments

On-Demand

- List Price
- No commitment for future ~~usage~~ ^{usage}

Reserved Instances (RIs)

- Traditional RIs, ~~also~~ Committed-Use Discounts (CUDs)
- Provide dedicated resources by agreeing to long-term usage of a specific machine type and region

~~Flexi~~ Savings plans

- Savings plans, Flexible CUDs
- A commitment to an aggregate spend amount
- No specified resource types, more flexible

Convertible RIs

- Like Traditional RIs, but have the added ^{flexibility} ~~cost~~ of being able to exchange for another resource

Enterprise Discount Plan (EDP)

- Long-term service agreements with a cloud service provider (CSP)
- Blanket ~~at~~ discount rate on all services ~~at~~
- Can include ~~it~~ Negotiated SKU discounts for identified services.

Bonus: Partner Discount

- Like an EDP, but ~~service agreement~~ ^{contract} is with a Partner / reseller that the company buys cloud services through.
- Blanket discount rate on all services

Strategies to

Get The Most Out of Cloud Commitments

Mix & Match

- no one-size-fits-all approach
- combine different commitment types to create a balanced and flexible cost structure.
- Tailor commitment types to specific needs of workloads.

Optimize Multi-year

- For stable workloads, consider longer-term, 3 year commitments which offer greater savings potential compared to shorter commits.

Convertible RIs

- Some service providers offer the option to exchange or modify a reservation to better align to changing workloads
- Adjust as needs change but offer much lower discount %.

Usage First

- Focusing on optimizing usage first can allow you to eliminate cloud waste so you only commit to what you need.

Discount Stacking

- Custom EDPs and Partner Discounts can stack on top of commitment savings, further increasing the savings potential.

Stagger Commitments

- Stagger the expiration timing for added flexibility

Third-Party tools & Services

- Streamline purchase, renewal, and management of commitments
- Automate based on pre-defined usage guardrails

Order of Operations:

How Commitment Discounts Are Applied

On-Demand Usage

- Baseline usage at list price.

Free Tier

- Automatically applies free tier first.

TRADITIONAL RIs /
Committed-Use discounts

Savings Plans /
Flexible CUDs

- Traditional, resource-based commits are applied first

- Savings plans, or spend-based commits, then operate across billing account on remaining on-demand usage.

Enterprise Discount
Plan (EDP)

- EDPs can stack on top of RIs, but not on top of negotiated SaaS discounts

PARTNER Discount

- Some companies will opt to purchase cloud services through a reseller.
- Discounts associated with this contract apply last before the invoice.

Invoice

- Final amount ~~is~~ billed on invoice.

include example
from next page →

example.

	<u>Core / hours</u>	<u>Unit Price</u>	<u>net Price</u>
on-Demand	100	\$10	\$1000
Free Tier	- 1	\$10	\$990
RIS	- 20	\$7	\$930
Savings plan	- 60	\$5	\$630
EDR	N/A	10% discount	\$567 invoice

~~100~~
~~- 1~~
~~99~~

~~cores~~
~~100~~
~~- 1 free~~
~~99 x 10~~